

ANALYSIS OVERVIEW

DATA SOURCES

- DuckDB table `catnat\_gaspar` (natural disasters in France, 1982–2025)
- DuckDB table `Res2\_INSEE`
(municipalities with secondary homes – years: 1968, 1975, 1982, 1990, 1999, 2011, 2016, 2022)
- Geographic source: <https://public.opendatasoft.com/api/explore/v2.1/catalog/datasets/georef-france-commune/exports/geojson>

HYPOTHESIS

Municipalities with a high proportion of secondary homes may be more exposed to certain natural risks.

OBJECTIVE

Determine whether municipalities with a high proportion of secondary homes are more frequently affected by natural disasters.

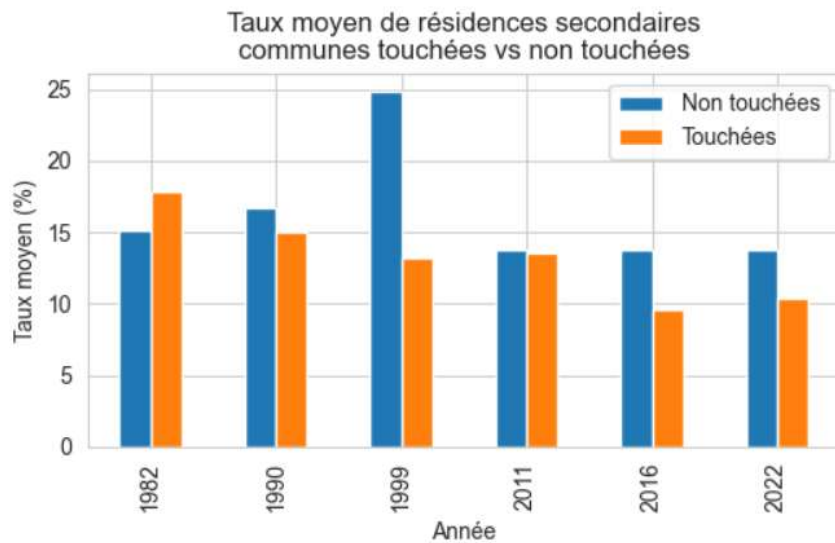
NOTE

Only years available in both datasets were analyzed: 1982, 1990, 1999, 2011, 2016, and 2022.

KEY FINDINGS

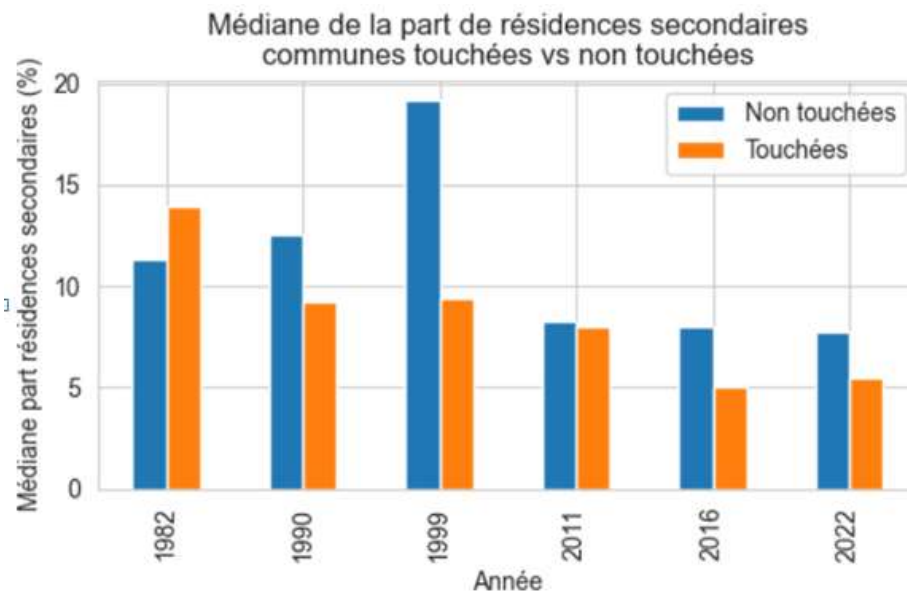
- No significant correlation between natural disaster exposure and the proportion of secondary homes.
- Highly exposed municipalities do not systematically have higher secondary home rates.
- Some disaster types (avalanches, landslides) are associated with high secondary home rates, but only in limited mountainous areas.
- The most frequent disasters (floods, ground movements) mainly affect municipalities with low secondary home rates (<15%).

✓ **Average rate of secondary homes in affected vs non-affected municipalities**



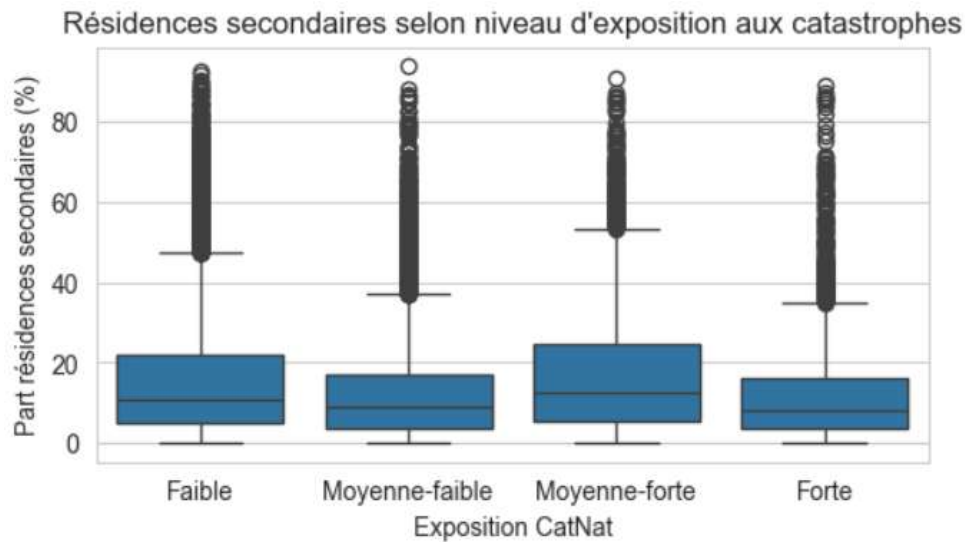
>>> The average proportion of secondary homes is similar between affected and non-affected municipalities, with no consistent pattern over time.

✓ **Median rate of secondary homes in affected vs. non-affected municipalities.**



>>> The median proportion of secondary homes shows no significant difference between affected and non-affected municipalities.

- ✓ **Rate of secondary homes by level of exposure to natural disasters.**



>>> The distribution of secondary home rates does not significantly vary across different levels of natural disaster exposure.

- ✓ **Top 10 municipalities most affected by natural disasters vs. average/median rate of secondary homes**

	codgeo	libgeo	total_catnat	mean_part_resid2	median_part_resid2
	28274 71574	Verzé	21	8.766667	7.80
	11770 31555	Toulouse	17	3.033333	3.00
	7882 24026	Bassillac et Auberoche	16	10.733333	10.55
	27742 71016	Azé	14	10.916667	9.95
	24562 63227	Mirefleurs	14	4.416667	2.35
	27454 70285	Héricourt	13	1.300000	1.20
	29753 76595	Saint-Jouin-Bruneval	13	10.816667	9.80
	29515 76349	Hautot-sur-Mer	13	12.033333	10.50
	29368 76192	Criel-sur-Mer	12	45.266667	43.70
	29427 76254	Étretat	12	39.866667	37.00

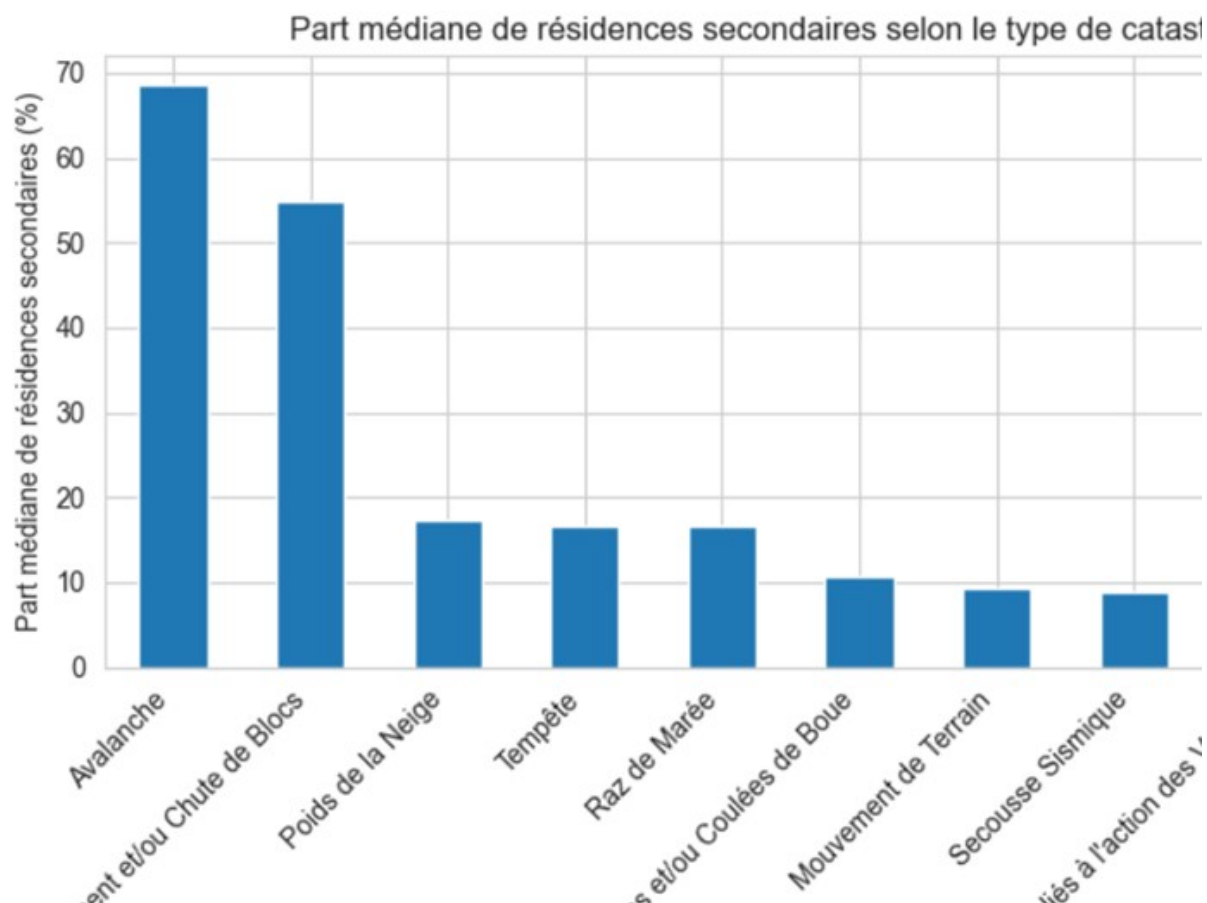
>>> The most affected municipalities do not systematically exhibit high proportions of secondary homes.

- ✓ **Top 10 municipalities with the highest proportion of secondary homes vs. number of natural disasters**

	codgeo	libgeo	total_catnat	mean_part_resid2	median_part_resid2	
	28905	73318	Villarembert	3	93.816667	93.85
	1798	05110	Puy-Saint-Vincent	0	92.533333	93.60
	1687	04226	Uvernet-Fours	0	91.766667	92.20
	21262	57579	Rhodes	4	90.800000	90.95
	1787	05098	Les Orres	0	90.200000	90.90
	25822	66004	Les Angles	2	89.816667	90.05
	1807	05119	Risoul	0	89.616667	90.70
	25001	64204	Eaux-Bonnes	6	89.216667	89.20
	1857	05177	Vars	0	88.883333	89.30
	1508	04006	Allos	0	88.583333	91.50

>>> *Municipalities with the highest proportions of secondary homes are not necessarily more exposed to natural disasters.*

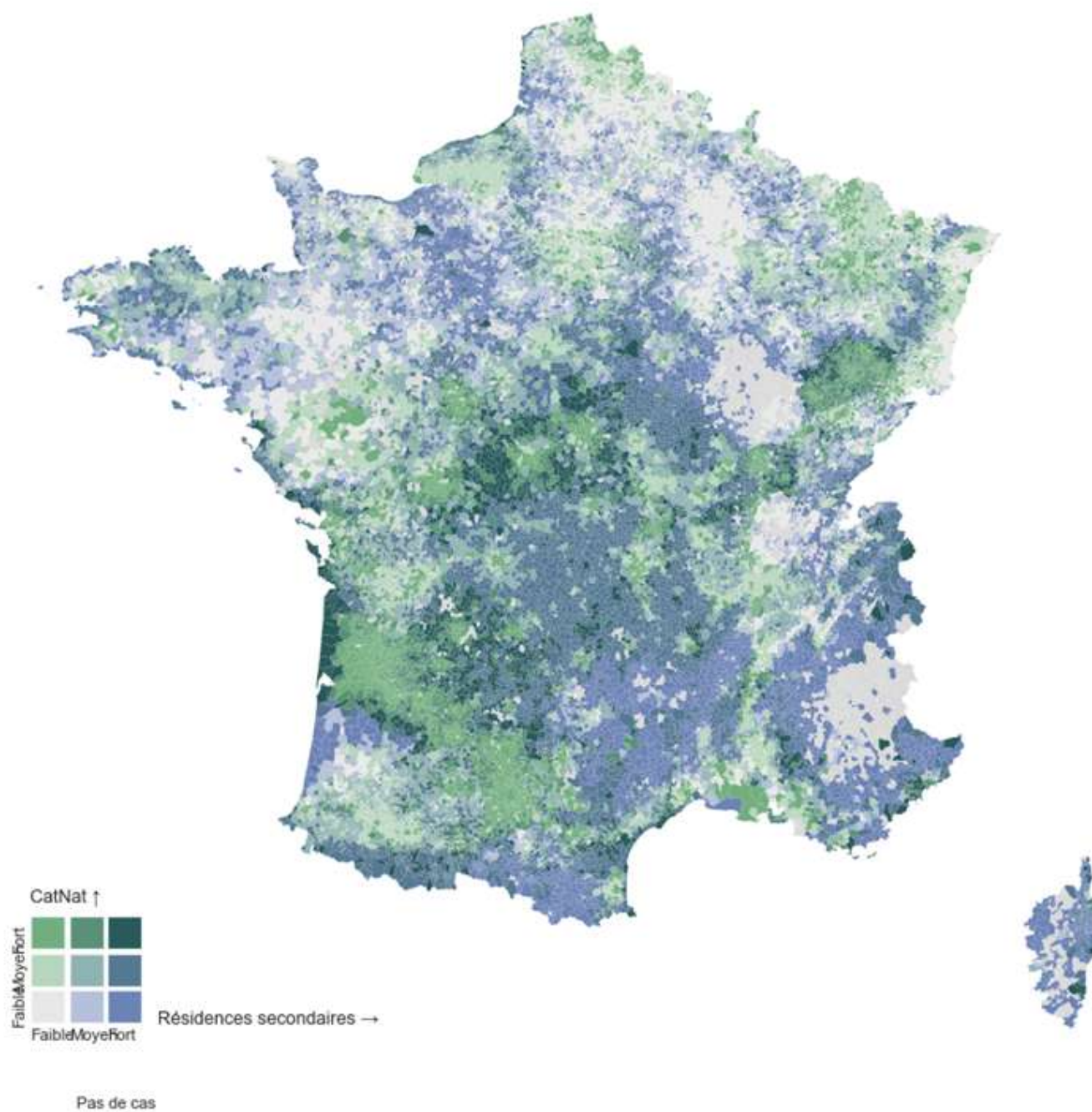
- ✓ **Average/median rate of secondary homes by type of natural disaster.**



	median_resid2	mean_resid2	nb_communes
lib_risque_jo			
Avalanche	68.50	67.033333	25
Eboulement et/ou Chute de Blocs	54.85	54.850000	2
Poids de la Neige	17.40	19.829294	1403
Tempête	16.70	20.382524	12074
Raz de Marée	16.60	16.600000	1
Inondations et/ou Coulées de Boue	10.60	14.849351	33647
Mouvement de Terrain	9.30	12.989343	26273
Secousse Sismique	8.90	12.570588	17

>>> *Avalanche- and landslide/rockfall-type disasters appear to be associated with municipalities that have a high proportion of secondary homes (67% and 55%, respectively). However, these results are based on a very limited number of municipalities and mainly reflect mountainous areas. In contrast, the most frequent natural disasters (floods and ground movements) predominantly affect municipalities where the share of secondary homes is below 15%, supporting the absence of an overall relationship between secondary homes and exposure to natural disasters.*

Catastrophes naturelles et résidences secondaires par commune



RESULTS

Several indicators were analyzed (means, medians, and segmentation by exposure quartiles) to assess both overall levels and distributions according to exposure intensity.

The differences observed between affected and non-affected municipalities are small and not systematic. Quartile-based analysis does not reveal any correlation between exposure level (low, medium, high) and the proportion of secondary homes.

The analysis of the 10 most extreme municipalities reinforces this observation: the most exposed municipalities do not necessarily have a high proportion of secondary homes, while some municipalities with a high share of secondary homes appear little or not affected.

Finally, analysis by type of natural disaster shows that certain risks, particularly avalanches and landslides, are associated with municipalities that have a high proportion of secondary homes. However, these results are based on a very limited number of cases and mainly reflect mountainous areas. In contrast, the most frequent natural disasters (floods and ground movements) affect municipalities with a low proportion of secondary homes (below 15%).

Bivariate mapping visually confirms these findings: areas highly exposed to natural disasters do not systematically correspond to territories with the highest proportion of secondary homes.

Overall, the hypothesis of a direct relationship between secondary homes and exposure to natural disasters is not supported by the available data.